

The Lorentz limiting ODE has finite support exactly off the diagonal

Candidate partial result for Conjecture 1 of arXiv:2303.04728

12 August 2026

Status. Candidate substantial partial result, likely valid; human review needed. The support-radius clause of the conjecture is proved, but the empirical-measure convergence and existence/uniqueness of the critical shooting solution remain open.

Source conjecture

Kabluchko, Prochno, and Sonleitner formulate Conjecture 1 on source PDF page 7 [1]:

For parameters $p > 1$, we currently do not have a probabilistic representation. However, we can use heuristic arguments based on maximum entropy considerations in order to derive the following conjecture about the limiting distribution.

Conjecture 1. *Let $1 \leq p \leq q < \infty$ and for each $n \in \mathbb{N}$ assume that $\tilde{X}^{(n)} = (\tilde{X}_1^{(n)}, \dots, \tilde{X}_n^{(n)})$ is uniformly distributed on $\tilde{\mathbb{D}}_{q,p}^n$. Then, for every bounded continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$, we have*

$$\frac{1}{n} \sum_{i=1}^n f(\tilde{X}_i^{(n)}) \xrightarrow[n \rightarrow \infty]{\mathbb{P}} \int_{\mathbb{R}} f(x) \nu_{q,p}(\mathrm{d}x),$$

where $\nu_{q,p}$ is a symmetric probability measure on $\mathbb{R}$ absolutely continuous with respect to Lebesgue measure and with density function $f_{q,p}: \mathbb{R} \rightarrow [0, \infty)$ satisfying $f_{q,p}(x) = \frac{1}{2} G'(|x|)$, where $G: [0, \infty) \rightarrow \mathbb{R}$ satisfies $G(0) = 0$ and is the unique solution to the differential equation

$$G''(x) = -G'(x)(1 - G(x))^{p/q-1} x^{p-1}, \quad x \in (0, r_{p,q}),$$

with $G(0) = 0$ and $\lim_{x \uparrow r_{p,q}} G'(x) = 0$, where $r_{p,q} \in (0, \infty]$ is the first point such that $G(r_{p,q}) = 1$. Further, we conjecture that $r_{p,q} = \infty$ if and only if $p = q$.

Varying the free parameter $G'(0) > 0$, we obtain different solutions, see Figure [1](#) for a simulation. If $G'(0)$ is smaller than some critical value $c_{p,q}$, then we conjecture that $\lim_{x \rightarrow \infty} G(x) < 1$. If $G'(0) = c_{p,q}$, then there is a (minimal) value $r_{p,q}$ such that $G(r_{p,q}) = 1$ and we have $\lim_{x \uparrow r_{p,q}} G'(x) = 0$. This critical solution is the one that appears in the above conjecture.

Remark 5. The conjecture encapsulates the case $p = q$ for ℓ_n^n balls, where it correctly returns p -

For $1 \leq p \leq q < \infty$, the conjectured positive-half distribution function G solves

$$G''(x) = -G'(x)(1 - G(x))^{p/q-1} x^{p-1}, \quad 0 < x < r_{p,q}, \tag{1}$$

with $G(0) = 0$, $G(r_{p,q}) = 1$, and $G'(x) \rightarrow 0$ as $x \uparrow r_{p,q}$. The authors explicitly conjecture that $r_{p,q} = \infty$ if and only if $p = q$.

Precise scope and proof intuition

Call (G, r) an *admissible boundary solution* if $r \in (0, \infty]$, G is a classical solution of (1) on $(0, r)$, $G(0) = 0$, $0 < G(x) < 1$, $G'(x) > 0$ there, and

$$\lim_{x \uparrow r} G(x) = 1, \quad \lim_{x \uparrow r} G'(x) = 0.$$

This is exactly the ODE-and-boundary portion of the conjectured critical solution. Positivity of G' also follows from a positive initial slope by the linear equation for G' as long as $G < 1$.

The key idea is to set $u = 1 - G$ and $w = G'$. After using the strictly decreasing quantity u as the independent variable, the ODE becomes

$$\frac{dw}{du} = x(u)^{p-1}u^{-\beta}, \quad \beta = 1 - \frac{p}{q}.$$

The terminal condition $w = 0$ makes this an exact integral formula. If $\beta > 0$ and the endpoint were infinite, future values of $x(u)$ would force w to be so large that u^β reaches zero in finite time. The same formula at the true finite endpoint determines the exact power with which the density vanishes.

Support dichotomy and edge law

Theorem 1. *Let $1 \leq p \leq q < \infty$ and let (G, r) be an admissible boundary solution of (1).*

1. *If $p = q$, then $r = \infty$ and necessarily*

$$G'(x) = a_p e^{-x^p/p}, \quad a_p = \left(\int_0^\infty e^{-t^p/p} dt \right)^{-1}. \quad (2)$$

2. *If $p < q$, then $r < \infty$. Put $\beta = 1 - p/q \in (0, 1)$. For every $0 < x < r$,*

$$\frac{\beta}{p(1-\beta)}(r^p - x^p) \leq (1 - G(x))^\beta \leq \frac{\beta r^{p-1}}{1-\beta}(r - x). \quad (3)$$

Moreover, as $x \uparrow r$,

$$1 - G(x) \sim \left(\frac{q-p}{p} r^{p-1} (r - x) \right)^{q/(q-p)}, \quad (4)$$

$$G'(x) \sim \frac{q}{p} r^{p-1} \left(\frac{q-p}{p} r^{p-1} (r - x) \right)^{p/(q-p)}. \quad (5)$$

Thus the conjectured density $G'(|x|)/2$ has the exact compact-support edge exponent $p/(q-p)$.

In particular, every admissible boundary solution satisfies

$$r = \infty \iff p = q.$$

Proof. Set

$$\beta = 1 - \frac{p}{q}, \quad u = 1 - G, \quad w = G'.$$

Then $u > 0$, $w > 0$, $u' = -w$, and (1) is equivalent to

$$w' = -x^{p-1}u^{-\beta}w. \quad (6)$$

Suppose first that $p = q$, so $\beta = 0$. Equation (6) integrates to $w(x) = ae^{-x^p/p}$ for some $a > 0$. It is positive at every finite point, so the boundary condition $w(x) \rightarrow 0$ cannot hold at a finite endpoint. Hence $r = \infty$. The condition $G(x) \rightarrow 1$ then fixes $a = a_p$ and proves (2).

Now assume $p < q$, so $0 < \beta < 1$. Since $u' = -w < 0$, u has an inverse on its range; write $x(v)$ for the unique point at which $u = v$. Dividing (6) by $u' = -w$ gives

$$\frac{dw}{du} = x(u)^{p-1}u^{-\beta}. \quad (7)$$

We first rule out $r = \infty$. If that were the endpoint, then $u \downarrow 0$ and $w \downarrow 0$. Integrating (7) from the terminal value $u = 0$ to the current value $u = u(x)$ gives

$$w(x) = \int_0^{u(x)} x(v)^{p-1} v^{-\beta} dv. \quad (8)$$

For $0 < v < u(x)$, the value v occurs later than $u(x)$, so $x(v) \geq x$. As $p \geq 1$, (8) implies

$$w(x) \geq \frac{x^{p-1}}{1-\beta} u(x)^{1-\beta}. \quad (9)$$

Consequently

$$\frac{d}{dx} u(x)^\beta = -\beta u(x)^{\beta-1} w(x) \leq -\frac{\beta}{1-\beta} x^{p-1}.$$

For a fixed $x_0 > 0$, integration to $X > x_0$ yields

$$u(X)^\beta \leq u(x_0)^\beta - \frac{\beta}{p(1-\beta)} (X^p - x_0^p),$$

whose right side is negative for large X . This contradiction proves $r < \infty$.

At the finite endpoint the same integration of (7) gives the exact identity

$$w(x) = \int_0^{u(x)} x(v)^{p-1} v^{-\beta} dv, \quad x < x(v) < r. \quad (10)$$

Therefore

$$\frac{x^{p-1}}{1-\beta} u(x)^{1-\beta} \leq w(x) \leq \frac{r^{p-1}}{1-\beta} u(x)^{1-\beta}. \quad (11)$$

Multiplying by $-\beta u^{\beta-1}$ reverses the order, and integrating from x to r gives exactly (3).

Finally, $x(v) \rightarrow r$ as $v \downarrow 0$. Divide (10) by $u(x)^{1-\beta}$ and use the elementary weighted-average limit to obtain

$$\frac{w(x)}{u(x)^{1-\beta}} \rightarrow \frac{r^{p-1}}{1-\beta}. \quad (12)$$

It follows that

$$\frac{d}{dx} u(x)^\beta \rightarrow -\frac{\beta r^{p-1}}{1-\beta}.$$

Since $u(r) = 0$, integration over the final interval gives

$$u(x)^\beta \sim \frac{\beta r^{p-1}}{1-\beta} (r-x).$$

Substituting $\beta = (q-p)/q$ proves (4); combining it with (12) proves (5). □

Corollary 2 (exact recovery of the known $p = 1$ case). *If $p = 1 < q$, every admissible boundary solution has*

$$r = \frac{1}{q-1}, \quad G'(0) = q,$$

and

$$G'(x) = q(1 - (q-1)x)^{1/(q-1)}, \quad 0 \leq x \leq \frac{1}{q-1}.$$

Proof. For $p = 1$, the factor $x(v)^{p-1}$ in (10) is identically one. Hence $w = u^{1-\beta}/(1-\beta) = qu^{1/q}$ exactly, so $(u^\beta)' = -(q-1)$. The conditions $u(0) = 1$ and $u(r) = 0$ give the stated formulas. This agrees with Theorem B of [1]. □

Verification, limitations, and novelty status

The diagonal formula recovers the usual normalized p -Gaussian, and the $p = 1$ corollary recovers both the density and support proved independently in the source paper. The two sides of (3) have the same first-order endpoint constant, providing a separate check on (4).

The theorem starts from exactly the ODE boundary-solution hypothesis already stated in Conjecture 1. It does *not* prove that the uniform measures on the Lorentz balls converge, nor that the critical initial slope exists or is unique. Focused shooting, entropy-variational, and Gibbs-conditioning upgrade attempts leave genuine compactness, regularity, exponential-tightness, and moving-constraint issues.

Cheap run indexes and bounded exact-title, arXiv-id, quoted-conjecture, $r_{p,q}$, support-radius, and general- p empirical-distribution searches through 12 August 2026 found the source and its published version but no later resolution of this clause. Novelty confidence is moderate.

Human-review recommendation. Check first the orientation $x(v) \geq x$ in (8), then the sign when integrating the differential inequality for u^β . Those two elementary steps carry the support dichotomy; the edge law is a direct endpoint limit of the same exact identity.

References

- [1] Z. Kabluchko, J. Prochno, and M. Sonnleitner, *A probabilistic approach to Lorentz balls*, arXiv:2303.04728 (2023), Conjecture 1, p. 7.